Choupo

The Open Source Chemical Process Simulator

Properties Guide

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Contents

"Thermodynamics is a funny subject. The first time you go through it, you don't understand it at all. The second time you go through it, you think you understand it, except for one or two points. The third time you go through it, you know you don't understand it, but by that time you are so used to it, it doesn't bother you any more."

— Arnold Sommerfeld

1 Introduction: the difficulty, the map, the promise

What this guide is — and is not. This is the props-**mode workflow** guide: how you *estimate* a missing compound, *fit* its binary pairs, and *consistency-test* the data — the see-then-decide loop of the Props/Explore workspace. It is *not* the property *theory.* For the first-principles derivations of NRTL, Wilson, UNIQUAC, UNIFAC, the cubic equations of state (SRK, Peng–Robinson), the electrolyte models (Pitzer, eNRTL, aqueous speciation), Henry's law, Rackett, Ambrose–Walton, sublimation and the NASA–CEA Gibbs-of-formation data, see the **Theory Guide** (the in-GUI model panels deep-link straight into it).

If thermodynamics felt slippery the first time — the formalism pristine, yet somehow never self-sufficient when you sat down with real data — you are in good company: Sommerfeld, who reshaped quantum physics, said he never wrote his thermodynamics book because he never felt he had understood the subject. The difficulty is real, but it is not formal incompleteness. It is that an *axiomatic-looking* macroscopic theory rests, underneath, on emergent statistical facts and on parameters that must be *measured*, not derived. The confusion is the seam between the two showing through. This guide does not hide that seam — it makes it the organising idea.

1.1 The target: where theory grips

Picture property prediction as a target (Figure 1): pristine at the core, increasingly empirical toward the rim.

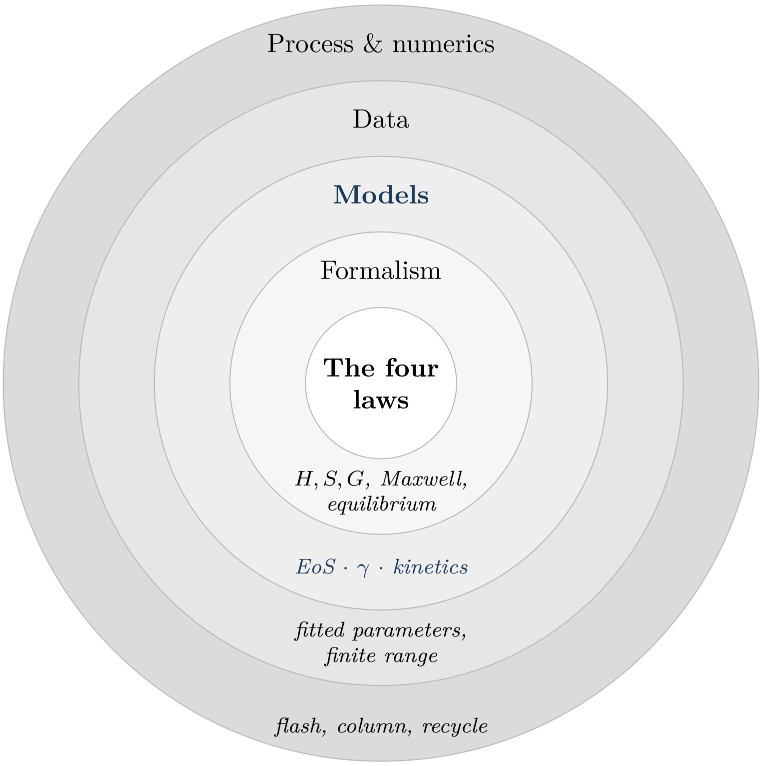

Figure 1: The layers of property prediction. The two inner shells follow from the laws alone. The accent shell — *constitutive models* — is the first fault line: the laws say *that* fugacities must match across phases, never *how* to compute one for a real mixture. From there outward, the theory is necessary but no longer sufficient.

- **The four laws (core).** The zeroth, first, second and third laws define temperature, internal energy, entropy, and — via Nernst — an absolute zero of entropy. Parameter-free.
- **Formalism.** Pure manipulation yields the potentials H, G, A, the Maxwell relations, the equilibrium criteria (equality of fugacity across phases, $\Delta G = -RT \ln K$ for reaction) and the phase rule. Still derivable from nothing but the laws.
- **Constitutive models — where it first grips.** The laws require fugacities to be equal across phases; they do not tell you how to *compute* a fugacity for a real mixture. You must supply a model they do not give you — an equation of state, an activity model, a kinetic law. In Choupo: `idealGas/SRK/PR`; `ideal/NRTL/Wilson/UNIFAC`.
- **Data — "go to the laboratory."** Those models carry parameters thermodynamics cannot predict: T_c, P_c, ω, the Antoine constants, $C_p^\circ(T)$, ΔH_f°, S°, the NRTL a_{ij}. Each is measured or estimated, and each has a finite

validity range. In Choupo, these are the `.dat` files.

- **Process & numerics.** Where the predicted numbers are consumed — flash, distillation, recycle — and grip a second time, now numerically (multiple EoS roots, phase stability, initialisation).

> **Key idea.** Sommerfeld's difficulty is this picture: a theory that *looks* axiomatic to the rim, but stops being self-sufficient at the accent shell. The honest move is not to paper over the seam between rigorous derivation and fitted correlation — it is to show exactly where it runs.

1.2 The glass-box promise

Because the outer shells are empirical, every number Choupo predicts should travel with two things attached:

- **Provenance** — which method produced it (rigorous derivation, fitted correlation, or group-contribution estimate) and the source of each parameter (a measured datum or an estimate).
- **Validity awareness** — whether a correlation is *extrapolating beyond its fitted range*, with a warning when it is, rather than a confident number where none is warranted.

> *Intuition.* Ask Choupo for the vapour pressure of nitrogen at room temperature and it *refuses*: N_2 is supercritical there (its critical temperature is −147 °C), and a saturation pressure does not exist above the critical point. The P^{sat} curve simply *breaks* at T_c instead of extrapolating Antoine into a meaningless ∼600 bar. That refusal is the whole promise in miniature: the model declines to invent a number the physics forbids.

1.3 Where we are heading

A single property over a wide temperature span is not one problem — it is several physical regimes stitched together, and your confidence in any predicted number rises and collapses as you cross them. Figure 2 is the destination of this guide: the vapour pressure of a **water / ethanol / benzene** mixture from −200 to 1000 °C, drawn not as a curve but as a band of *confidence*.

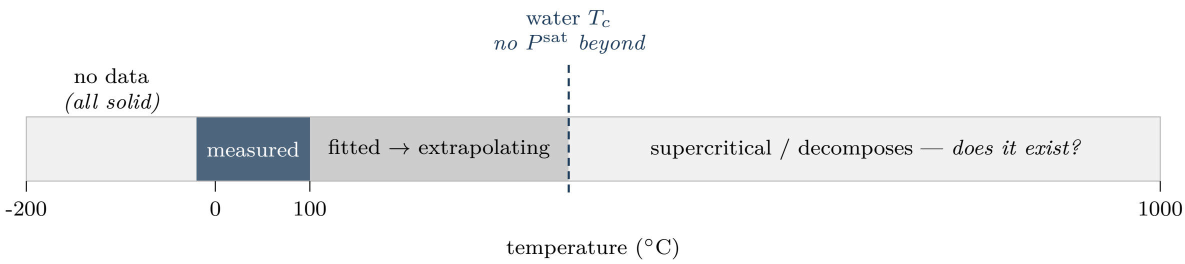

Figure 2: The same property, five honesties. A narrow window where we actually have data; degradation outward as the correlation extrapolates; a hard *wall* at the critical temperature where the property ceases to exist (the nitrogen refusal, generalised); and a cryogenic desert where every species has frozen solid (ethanol last, at −114 °C) and there is simply no data. The theory runs to both edges; reality and data do not.

We build each piece of that picture in the chapters that follow — estimating a missing component (§3), curating the mixture models against lab points (§4), and testing whether the data itself is even consistent (§5). We read that span regime by regime at the end of this introduction (§1.6), and return to solve it numerically, end to end, at the close.

1.4 Anatomy of a component: where the numbers come from

Open any `.dat` and you see a flat list of numbers. The natural question — the one this whole guide turns on — is: *which of these are independent inputs, and which derived properties does the engine actually build from them?* Let us read `acetone.dat` field by field and follow each number to the property it feeds and to the line of code that consumes it.

Field	What it is, and what it builds
Identity	
`MW`	molar mass; converts molar ↔ mass everywhere.
Caloric anchors — `gibbsFormation`	
`dHf_298` ΔH_f°	the *position* of the enthalpy scale: bond energy C_p knows nothing about (combustion calorimetry).
`s_298` S°	the *absolute* (third-law) entropy at 298 K — a true zero-referenced value, not a difference.
Caloric shape — `idealGasHeatCapacity`	
`coefficients` $C_p^\circ(T)$	the master correlation: integrated to give the *temperature shape* of H, S, G (below).
Corresponding-states trio	
`Tc`, `Pc`	critical point: set the *size* of the cubic-EoS attraction term a_c, and the reduced temperature $T_r = T/T_c$.
`omega` ω	acentric factor: sets the *temperature shape* of that attraction, $\alpha(T_r;\omega)$ — how non-spherical / polar the molecule is.
Phase change	
`Tb`, `HvapTb`	boiling point + latent heat at T_b; with `Tc` drive Watson's $\Delta H_{\mathrm{vap}}(T)$.
`Vliq`	liquid molar volume (Poynting, liquid density).
Correlations & recipes	
`vaporPressure`	Antoine $\log_{10} P^{\mathrm{sat}} = A - B/(T+C)$, with its `Trange`; the saturation curve.
`groups`	the molecular decomposition a group-contribution estimator / UNIFAC reads (the curation recipe, §3).
`diffusionVolume`	Fuller gas diffusivity ⇒ Lewis number (wet-bulb).

The caloric engine: H, S, G from C_p

This is the mechanism behind "other properties that derive from here" — and it is worth seeing *how one arrives at* $S(T)$, since that is the step that trips everyone up. Start from the laws, at constant pressure. Heat capacity *is* the temperature slope of enthalpy, $C_p = (\partial H/\partial T)_P$, so

$$dH = C_p\, dT.$$

For entropy, the second law gives $dS = \delta q_{\mathrm{rev}}/T$; at constant pressure the reversible heat is $\delta q_{\mathrm{rev}} = C_p\, dT$, hence

$$dS = \frac{C_p}{T}\, dT.$$

That single step — dividing by T — is the whole origin of the $\int C_p/T$. Integrate each from a reference temperature and add the 298 K anchor; Choupo stores $C_p^\circ(T)$ and the two anchors (*not* tables of H, S) and evaluates:

$$H^\circ(T) = \underbrace{\Delta H_f^\circ}_{\text{position}} + \int_{298.15}^{T} C_p^\circ(T')\, dT' \qquad (\texttt{Component::h\_pure\_ig}, \texttt{Component.cpp}) \tag{1}$$

$$S^\circ(T) = \underbrace{S^\circ_{298}}_{\text{absolute}} + \int_{298.15}^{T} \frac{C_p^\circ(T')}{T'}\, dT' \qquad (\texttt{Component::s\_pure\_ig}) \tag{2}$$

$$G^\circ(T) = H^\circ(T) - T\, S^\circ(T) \tag{3}$$

For a mixture the engine adds the entropy of mixing $-R\sum y_i \ln y_i$ and the pressure term $-R\ln(P/P_{\mathrm{ref}})$ (`ThermoPackage::S_ig`). That is the *entire* ideal-gas caloric model — no hidden tables.

These integrals are not done numerically. For a polynomial C_p the primitive is elementary — it was solved *once, on paper* — so each call evaluates a closed form,

$$\int_{T_{ref}}^{T} \frac{C_p^\circ}{T'}\, dT' = a_0 \ln \frac{T}{T_{ref}} + \sum_{k\geq 1} \frac{a_k}{k}\left(T^k - T_{ref}^k\right),$$

exact to machine precision in a handful of operations (`PolynomialCp::S`). The standard correlations all share this property (NASA, Aly–Lee, Shomate have closed primitives too); Choupo never runs a quadrature loop for C_p.

> **Worked example 1.1.** Acetone H and S at 400 K — by hand, then by ChoupoWith $C_p^\circ(T) = 6.301 + 0.2606\,T - 1.253\times10^{-4}T^2 + 2.038\times10^{-8}T^3$ (units J mol^{-1}K^{-1}), integrating from 298.15 to 400 K:
>
> $$\int C_p^\circ\, dT' = 8431 \text{ J/mol}, \qquad \int \frac{C_p^\circ}{T'}\, dT' = 24.19 \text{ J/(mol}\cdot\text{K)}.$$

So $H^\circ(400) = -217100 + 8431 = \mathbf{-208669}$ J/mol and $S^\circ(400) = 295.30 + 24.19 = \mathbf{319.49}$ J/(mol · K). Running `choupoProps` on a one-point scan returns exactly $H_{\text{ig}} = -208669$ J/mol, $S_{\text{ig}} = 319.49$ J/(mol · K). The number on the page is the number in the code — and at 298.15 K both integrals vanish, so $H = \Delta H_f^\circ$ and $S = S^\circ_{298}$, the anchors laid bare.

Key idea. C_p gives the *shape*; the formation values give the *position.* That is why the `.dat` lists $C_p^\circ(T)$ *and* ΔH_f°, S° separately — not redundancy. And note the asymmetry: entropy has an absolute zero for free (third law: $S \to 0$ for a perfect crystal at $T = 0$), so S°_{298} is a genuine absolute; enthalpy has *no* natural zero, so ΔH_f° must be *measured* and supplied.

What the critical properties and acentric factor actually do

T_c, P_c, ω do nothing in the caloric equations above — they enter only when you ask for *real-fluid* behaviour through a cubic equation of state. There the attraction term is $a(T) = a_c\,\alpha(T_r)$, with a_c fixed by T_c, P_c and the temperature shape fixed by ω (Soave/SRK):

$$\alpha(T) = \left[\,1 + m(1 - \sqrt{T_r})\,\right]^2, \quad m = 0.48508 + 1.55171\,\omega - 0.15613\,\omega^2, \quad T_r = \tfrac{T}{T_c}$$

(`SRK.cpp`; Peng–Robinson uses the analogous $\kappa(\omega)$). This $a(T)$ is what produces the departure functions $H - H^\circ$, $S - S^\circ$ and the compressibility Z — the real-gas correction layered on top of the ideal-gas caloric result. The same T_c, T_b also drive Watson's latent heat, $\Delta H_{\text{vap}}(T) = \Delta H_{\text{vap}}(T_b)\,[(T_c - T)/(T_c - T_b)]^{0.38}$ (`Component.cpp`).

> *Intuition.* So why does `acetone.dat` carry ω at all if you are running the default `idealGas`? Because the data is *model-agnostic* and the model decides which fields wake up. Under `idealGas` there is no departure, and ω, T_c, P_c are simply never read. Select `SRK` for a high-pressure compressor and the very same numbers come alive. The `.dat` is the laboratory record; the model is the question you ask of it.

1.5 From G to equilibrium: what ΔG° is (and is not)

The single most common confusion in this whole subject is ΔG° versus ΔG. It is worth settling once, because G is what the reactors minimise and it is built from exactly the H and S above.

First, the $^\circ$. It means *reference state* — each species pure, ideal gas, at 1 bar — but *at the reaction temperature* T. It does *not* mean "at 298 K." So $\Delta G^\circ = \Delta G^\circ(T)$ varies with temperature; $\Delta G^\circ(298)$ is just one point on a curve.

Then, the two are not the same thing.

- ΔG° is the Gibbs change if reactants $\to$ products with *everything in its standard state.* It is a *fixed number* — a property of the reaction at T.
- ΔG is the *actual* Gibbs change at the real composition (the real partial pressures). It is the *driving force.*

The reaction quotient Q bridges them, and equilibrium is where the driving force — not the reference number — vanishes:

$$\Delta G = \Delta G^\circ + RT \ln Q, \qquad \Delta G = 0 \text{ at equilibrium } \Rightarrow \boxed{\Delta G^\circ = -RT \ln K}$$

Key idea. At equilibrium it is ΔG that is zero, *never* ΔG°. ΔG° stays fixed and merely sets *where* the equilibrium sits (the value of K); the system gets there by shifting Q until $\Delta G = 0$. And $\Delta G^\circ > 0$ does not mean "no reaction" — it means $K < 1$, i.e. partial conversion. Only the sign and size of K change.

Where the number comes from — straight back to the caloric engine:

$$\Delta G^\circ(T) = \Delta H^\circ(T) - T\,\Delta S^\circ(T) = \sum_i \nu_i\, g_i^\circ(T), \qquad g_i^\circ(T) = h_i^{\text{ig}}(T) - T\, s_i^{\text{ig}}(T).$$

Note the subtlety that confuses everyone: h^{ig} carries the *formation* enthalpy (ΔH_f°, an elements datum) while s^{ig} is the *absolute* (third-law) entropy. Mixing the two references looks wrong — but it is exactly right, because in a *balanced* reaction the element contributions cancel through $\sum_i \nu_i$. That is precisely why the `.dat` carries ΔH_f° *and* S° separately: each plays a different role, and they combine correctly only across a balanced stoichiometry (`Component::g_pure_ig`; consumed in `PFR.cpp`/`CSTR.cpp` as $K = \exp(-\Delta G^\circ/RT)$ and in the Gibbs reactor's direct G-minimisation).

How K moves with T (Gibbs–Helmholtz). $\Delta G^\circ(T)$ is not a frozen datum — it has its own temperature law, and that law is a workhorse relation worth knowing by name. Gibbs–Helmholtz states $[\partial(G/T)/\partial T]_P = -H/T^2$; applied to $\Delta G^\circ = -RT\ln K$ it collapses to the van 't Hoff equation,

$$\boxed{\frac{d\ln K}{dT} = \frac{\Delta H^\circ}{RT^2}}$$

which is just Le Chatelier made quantitative: an exothermic reaction ($\Delta H^\circ < 0$) has K *falling* as T rises. In Choupo it doubles as a glass-box self-check — the $K(T)$ the reactors build from $g^\circ(T)$ at each temperature must show exactly this slope, or the enthalpy and the equilibrium constant are disagreeing with each other.

> *Intuition.* These relations are not ornaments. Gibbs–Helmholtz is *how the equilibrium moves with temperature* (above); its sibling, the **Gibbs–Duhem** equation $\sum_i x_i \,\mathrm{d}\ln\gamma_i = 0$ (constant T, P), says the activity coefficients of a mixture are *not independent.* That single constraint is exactly why measured VLE data can be *tested* for thermodynamic consistency — and why an activity model is built from one G^{E} function rather than fitting each γ_i freely. It is the engine of §5.

1.6 A worked span: water / ethanol / benzene, -200 to 1000 °C

Put the three together and sweep the temperature. This one mixture crosses *every* regime of property prediction, and is the honest way to see both how properties are obtained *and* where the models give out — which is exactly the colour band of Figure 2, now read regime by regime. (Critical temperatures: ethanol 241 °C, benzene 289 °C, water 374 °C; melting points: ethanol −114, water 0, benzene 5.5 °C.)

Regime	How a property is obtained	Where the model gives out
-200 to ~ 0 °C solids (SLE)	melting points T_m, ΔH_{fus}, solid C_p; freezing-point depression; sublimation P^{sat}	**low-T edge.** Below every Antoine `Trange`: P^{sat} would extrapolate blindly. Solid-phase data is sparse — a cryogenic *data desert.*
~ 0 to ~ 65 °C two liquids (LLE)	activity model (`NRTL`/`UNIFAC`) for γ; water/benzene nearly immiscible, ethanol the cosolvent	VLE-fitted parameters do *not* serve LLE; `Wilson` cannot represent a liquid split at all; without T-dependent parameters the miscibility gap appears/vanishes wrongly.
~ 65 to ~ 100 °C boiling (VLLE)	Antoine P^{sat} + γ–ϕ + three-phase flash (Gibbs minimisation); ethanol/water + ternary azeotrope (~ 64.9 °C)	azeotrope sensitivity — a small γ error moves the column design; bubble-point (Wang–Henke) distillation is unstable *through* an azeotrope (use `method simultaneous`).
~ 100 to ~ 290 °C vapour → near-critical	cubic EoS (`SRK`/`PR`) with k_{ij}; departure functions $H - H^\circ$, $S - S^\circ$, Z	cubics degrade as the critical region is approached; k_{ij} often needs to be made temperature-dependent over so wide a span.
~ 290 to 374 °C mixed super/sub-critical	the method itself must switch γ–$\phi \to \phi$–ϕ (one EoS for every phase)	above a species' T_c the activity-coefficient picture loses meaning; no single γ model spans the switch; near-critical accuracy is poor.
374 to 1000 °C supercritical + reaction	ϕ–ϕ EoS; wide-range $C_p^\circ(T)$ for the caloric terms	**high-T edge.** P^{sat} ceases to exist above T_c (Choupo *refuses* it, §1.2); and the species thermally *decompose* — the fixed-component assumption fails: do they still exist?

The two edges — the ones you asked to watch — fail for opposite reasons but teach the same lesson. At the **cold edge** there is simply no data: every correlation is fitted to a window that starts well above 73 K, the species are frozen, and a confident number there would be invention, not prediction. At the **hot edge** the theory keeps producing numbers — Antoine will gladly report a vapour pressure at 800 °C — but the *physics* has moved on: past each T_c no saturation exists, and past a few hundred degrees the molecules are no longer the molecules in the `.dat`. In both cases the correct output is a *flag*, not a figure.

> **Common pitfall.** No single model spans this axis. A run that quietly uses one γ model and one cubic from −200 to 1000 °C is not "general" — it is extrapolating through three regime changes in silence. The wide span is not one problem; it is five problems wearing one temperature axis.

> *Intuition.* What Choupo ships *today* covers the middle of this span honestly: cubic EoS, `ideal`/`NRTL`/`Wilson`/`UNIFAC` activity, Antoine P^{sat}, Watson ΔH_{vap}, three-phase VLLE by Gibbs minimisation, and azeotropic columns via the simultaneous MESH solver. It does *not* (yet) carry association models

> (PC-SAFT/CPA) for the hydrogen-bonding non-ideality, nor electrolyte or wide-range near-critical packages. So for the H-bonding mid-span and the two edges, Choupo's job is not to fake accuracy it does not have — it is to predict where it can and to *say so* where it cannot. That honesty is the product.

2 Why a separate guide

The Theory Guide answers *how a model works* — how a flash splits, how an EoS predicts Z, how Levenberg–Marquardt minimises a residual. This guide answers a question that comes *before* that: *where do the numbers that feed the models come from?* A simulation is only as good as the T_c, the NRTL pair, the heat of formation it stands on, and those are not handed down — they are estimated, fitted, and tested.

Choupo treats that as a first-class phase with its own binary (`choupoProps`) and its own lifecycle:

> **CREATE** a component (from groups) → **CURATE** the mixture models (fit pairs, test data, choose by evidence) → **WIRE** the flowsheet → **SIMULATE**.

A `propsDict` of `operations` drives the first two stages. A case may have *no* `flowsheetDict` at all (a pure properties case — the compound bench), or carry one and use props to curate the thermo it will simulate. Same files, same engine; only the centre of the GUI differs.

> *Intuition.* The usual reflex is to click a property method and trust a badge. Choupo's stance (the "see, then decide" principle) is the opposite: you make the estimate or the fit *visible* — the group build-up, the model curve against the lab points, the consistency residual — and decide from the evidence in front of you. Estimation has error; the point is to see it.

3 Creating a component: Joback group contribution

When a component is missing from the catalogue and from the literature, you can *estimate* its pure-component constants from its molecular structure — the groups it is built from. Choupo implements the **Joback** method (Joback & Reid, 1987; tabulated in Poling, Prausnitz & O'Connell, *The Properties of Gases and Liquids*, 5th ed.), the `estimateComponent` property operation.

3.1 The method

Each first-order group contributes a tabulated increment. Summing the increments and substituting into Joback's correlations gives the constants (with n_A the total number of atoms):

$$T_b = 198.2 + \sum_i n_i\,\Delta T_{b,i} \tag{4}$$
$$T_c = T_b \Big/ \left(0.584 + 0.965\,\Sigma\Delta T_c - (\Sigma\Delta T_c)^2\right) \tag{5}$$
$$P_c = \left(0.113 + 0.0032\,n_A - \Sigma\Delta P_c\right)^{-2} \quad \text{[bar]} \tag{6}$$
$$V_c = 17.5 + \sum_i n_i\,\Delta V_{c,i} \quad [\mathrm{cm^3/mol}] \tag{7}$$
$$\Delta H_f^\circ(298) = 68.29 + \sum_i n_i\,\Delta H_{f,i} \quad \text{[kJ/mol, ideal gas]} \tag{8}$$
$$\Delta H_{\mathrm{vap}}(T_b) = 15.30 + \sum_i n_i\,\Delta H_{v,i} \tag{9}$$
$$C_p^{\mathrm{ig}}(T) = (\Sigma a - 37.93) + (\Sigma b + 0.210)T + (\Sigma c - 3.91\times10^{-4})T^2 + (\Sigma d + 2.06\times10^{-7})T^3. \tag{10}$$

The acentric factor ω is *not* a group sum; Choupo derives it from (T_b, T_c, P_c) by the Lee–Kesler correlation, with $T_{br} = T_b/T_c$ and P_c in atm:

$$\omega = \frac{-\ln P_c - 5.97214 + 6.09648/T_{br} + 1.28862\ln T_{br} - 0.169347\,T_{br}^6}{15.2518 - 15.6875/T_{br} - 13.4721\ln T_{br} + 0.43577\,T_{br}^6}. \tag{11}$$

A few first-order group increments (Joback & Reid; the full table is in the code, `src/propertyOps/EstimateComponent.cpp`):

group	ΔT_b	ΔT_c	ΔP_c	ΔH_f (kJ/mol)
$-CH_3$	23.58	0.0141	−0.0012	−76.45
$>CH_2$	22.88	0.0189	0.0000	−20.64
−OH (alcohol)	92.88	0.0741	0.0112	−208.04
>C=O (ketone)	76.75	0.0380	0.0031	−133.22
aromatic =CH−	26.73	0.0082	0.0011	2.09

3.2 Worked example: acetone

Acetone, $CH_3-CO-CH_3$, decomposes into $2\times$ `CH3` and $1\times$ `ketone`. The operation prints the build-up and validates against a `reference` block:

Property	Joback	Reference (NIST)	Dev.
T_b	322.1 K	329.2 K	−2.2%
T_c	500.6 K	508.1 K	−1.5%
P_c	48.0 bar	47.0 bar	+2.2%
V_c	209.5 cm^3/mol	209 cm^3/mol	≈ 0
ω	0.295	0.307	−3.9%
ΔH_f°	−217.8 kJ/mol	−217.1 kJ/mol	+0.3%

Within a few percent on the criticals and almost exact on ΔH_f — and you *see* the error, because the reference sits beside the estimate.

> *Intuition.* **Know the method's limits.** Joback is weak on T_b of strongly hydrogen-bonding species (ethanol's T_b comes out ∼14 K low), and it yields *no* Antoine / vapour-pressure parameters — those need a fit (§4) or a corresponding-states correlation, a separate step. An estimate is a starting point you then anchor to data, never a final answer.

> **Runnable case.** `tutorials/props/estimate/estimate_acetone` — the example above, groups in, validated estimate out.

4 Curating mixture models: fitting binary pairs

A pure-component set is not enough for a mixture: the non-ideality lives in the *binary interactions* (NRTL, Wilson). Those parameters are regressed against experimental VLE data by Levenberg–Marquardt — the `fitParameters` operation.

4.1 The activity-coefficient models

For a binary, the **NRTL** model gives the liquid-phase activity coefficients from three adjustable numbers per pair — τ_{12}, τ_{21} (temperature-dependent) and the non-randomness α:

$$\tau_{ij} = a_{ij} + \frac{b_{ij}}{T}, \qquad G_{ij} = \exp(-\alpha\,\tau_{ij}), \tag{12}$$

$$\ln\gamma_1 = x_2^2\left[\tau_{21}\left(\frac{G_{21}}{x_1 + x_2 G_{21}}\right)^2 + \frac{\tau_{12}\,G_{12}}{(x_2 + x_1 G_{12})^2}\right], \tag{13}$$

and $\ln\gamma_2$ by swapping the indices. So a pair carries four fittable coefficients $(a_{12}, b_{12}, a_{21}, b_{21})$ plus a fixed α (often 0.3). **Wilson** is the two-parameter alternative,

$$\ln\gamma_1 = -\ln(x_1 + \Lambda_{12}x_2) + x_2\left[\frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{x_2 + \Lambda_{21}x_1}\right], \tag{14}$$

which cannot represent a liquid–liquid split (use NRTL when one is suspected).

4.2 The residual the fit minimises

The measurable is the bubble temperature. At a fixed pressure P and liquid composition x, the bubble point is the T that closes modified Raoult,

$$\sum_i x_i\,\gamma_i(x,T)\,P_i^{\mathrm{sat}}(T) = P \tag{15}$$

(a Newton-1D in T; P_i^{sat} from Antoine). Levenberg–Marquardt then drives the pair coefficients $\theta = (a_{12}, b_{12}, a_{21}, b_{21})$ to minimise the sum of squared bubble-temperature residuals against the data,

$$\min_\theta \sum_k \left[T_{\mathrm{bub}}(x_k;\theta) - T_k^{\mathrm{exp}}\right]^2. \tag{16}$$

> *Intuition.* **A low residual is not a good fit.** At a *single* pressure $\tau_{ij} = a_{ij} + b_{ij}/T$ is nearly linear in both a_{ij} and b_{ij}/T over the narrow T range of one isotherm/isobar, so a and b are *correlated* — many (a, b) pairs fit equally well and the optimum is a valley, not a point. `fitParameters` forms $J^{\mathsf{T}}J$ at the optimum and reports the parameter correlation + confidence intervals so you SEE this; the Theory Guide's *Levenberg–Marquardt* and *What is fittable* chapters derive the diagnostic. Fit across pressures, or fix b, to break it.

4.3 See, then decide: the model overlay

Before fitting, look. Declare one scan `operation` per candidate model (ideal, NRTL, Wilson), emitting the $(x, T_{\mathrm{bubble}}, y_{\mathrm{eq}})$ trio, and an `experimental {}` entry that overlays them — and, with a `dataset`, draws the measured points on top. The student sees which model the data actually picks: for ethanol/water at 1 atm the ideal curve misses the azeotrope entirely while NRTL lands on it.

> **Runnable case.** `tutorials/props/compare/compare_vle_etoh_water` — ideal vs. NRTL bubble/dew curves overlaid with measured VLE points (Carey & Lewis, 1932).

4.4 Fit, then promote

When a model earns its keep, `fitParameters` regresses its coefficients to the data and writes a proposal to `constant/binaryPairs/<model>/<pair>.fit-<date>.dat`. Promotion is a deliberate, visible act: you inspect the parity plot and the identifiability statistics (confidence intervals, parameter correlation) the fit reports — never a "good fit" badge — and copy the proposal into the case's `constant/` when you trust it.

5 Testing the data itself: consistency

Before fitting a model *to* data, test whether the data is even thermodynamically consistent — a fit to bad data is a polished wrong answer. The `vleConsistency` operation does this in three steps.

5.1 Step 1 — activity coefficients straight from the data

It does not assume a model. From each measured (x_1, y_1, T) at the run pressure P, modified Raoult gives the activity coefficient of each component directly:

$$\gamma_i = \frac{y_i\,P}{x_i\,P_i^{\mathrm{sat}}(T)}, \qquad P_i^{\mathrm{sat}}(T) \text{ from Antoine.} \tag{17}$$

These are *experimental* γ_i — the test now asks whether they obey thermodynamics.

5.2 Step 2 — the Gibbs–Duhem point test

The Gibbs–Duhem equation is the constraint every real mixture must satisfy at constant T, P:

$$\sum_i x_i\,\mathrm{d}\ln\gamma_i = 0 \quad\Longrightarrow\quad x_1\frac{\mathrm{d}\ln\gamma_1}{\mathrm{d}x_1} + x_2\frac{\mathrm{d}\ln\gamma_2}{\mathrm{d}x_1} = 0 \;\; \text{(binary).} \tag{18}$$

The operation evaluates the left-hand side at each interior point (central differences in x_1) and reports its max and mean magnitude — the *point residual*. A consistent set keeps it near zero. (Isobaric data adds a small $-H^{\mathrm{E}}/RT^2 \cdot \mathrm{d}T/\mathrm{d}x$ term, dropped here; stated as a caveat, not hidden.)

5.3 Step 3 — the Herington area test

Integrate (18) across the whole composition range: for an isothermal set Gibbs–Duhem forces the net area of $\ln(\gamma_1/\gamma_2)$ to vanish. Herington's isobaric form compares that net area to the absolute area and corrects for the boiling-range:

$$D = 100\,\frac{\left|\displaystyle\int_0^1 \ln\frac{\gamma_1}{\gamma_2}\,\mathrm{d}x_1\right|}{\displaystyle\int_0^1 \left|\ln\frac{\gamma_1}{\gamma_2}\right|\mathrm{d}x_1}, \qquad J = 150\,\frac{T_{\max} - T_{\min}}{T_{\min}}, \tag{19}$$

(both integrals by the trapezoid rule over the sorted points). The set **passes** when $|D - J| < 10$. The GUI's consistency card plots $\ln\gamma_1$, $\ln\gamma_2$ and $\ln(\gamma_1/\gamma_2)$ vs x_1 with the net area shaded — you SEE the ± areas cancel, not just the number.

Finally the infinite-dilution coefficients $\gamma_1^\infty, \gamma_2^\infty$ (the dilute-end limits) are reported — the most stringent point of any model later fitted to the set.

> **Runnable case.** `tutorials/props/compare/compare_vle_etoh_water` — the `vleConsistency` op on the Carey–Lewis ethanol/water data (Herington $|D - J| = 4.6 < 10$, PASS).

This guide grows with the props pillar. Planned: a second group-contribution method (Constantinou–Gani) to compare against Joback; group contribution for liquid viscosity; a structure (SMILES) front-end for the compound bench.